



Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • goal: uncover spending patterns, segments, preferences, subscriptions

Dataset Snapshot

Volume

Rows: 3,900

Features

18 columns –
demographics,
purchase details,
behavior

Missing

Review Rating
missing: 37
values



Data Preparation (Python)



01

Load & Explore

pandas, df.info(), describe()

02

Impute Ratings

Median per product category

03

Standardize

rename → snake_case

04

Feature Eng.

age_group, purchase_frequency_days

05

Drop Redundancy

promo_code_used removed (redundant with discount_applied)

Database Integration



Cleaned DataFrame loaded to PostgreSQL for SQL analysis and business queries.

Why SQL?

Structured queries,
aggregations, segmentation

Outcome

Answer business questions at
scale

Key SQL Findings — Revenue & Spending



Revenue by Gender

Female: 75,191 • Male:
157,890



High-Spend Discount Users

839 rows – customers used discounts but spent above average



Top Rated Items

Gloves 3.86 • Sandals 3.84
• Boots 3.82 • Hat 3.80 •
Skirt 3.78

Shipping, Subscriptions & Repeat Behavior



- Shipping avg spend – Standard: 58.46 • Express: 60.48
- Subscribers: 1,053 customers – avg spend 59.49 – total revenue 62,645
- Non-subscribers: 2,847 – avg spend 59.87 – total revenue 170,436
- Repeat buyers (>5 purchases): more likely to subscribe (data shown)

Product & Discount Insights



Discount-Dependent

Top items with high discount rates: Hat 50% • Sneakers 49.66% • Coat 49.07% • Sweater 48.17% • Pants 47.37%



Top Products by Category

Accessories: Jewelry, Sunglasses, Belt – Clothing: Blouse, Pants, Shirt – Footwear: Sandals, Shoes, Sneakers – Outerwear: Jacket, Coat

Customer Segments & Age Revenue



Segments

Loyal: 3,116 • Returning: 701 •
New: 83

Revenue by Age Group

Young Adult 62,143 • Middle-aged
59,197 • Adult 55,978 • Senior
55,763



Power BI Dashboard Highlights

3.9K customers • Avg purchase \$59.76 • Avg rating 3.75 •
Subscriptions: Yes 27% / No 73%

Top revenue: Clothing (~90K) • Sales: Clothing (~1,800)

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow "Loyal" segment

Review Discount Policy

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users